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REM Sleep Parasympathetic Activity Predicts Wake Hyperarousal Symptoms Following a Traumatic Event

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Abstract

Heart rate variability (HRV) can be used to assess changes in the parasympathetic nervous system (PNS). Considering that patients with PTSD often experience disturbances in sleep, arousal, and autonomic functioning, we sought to explore the association of PNS activity during sleep with hyperarousal symptoms of PTSD. Because a broad literature supports the importance of REM sleep in PTSD, REM-sleep features were specifically examined as predictors of PTSD symptom severity. 90 participants, primarily civilian and female, aged 18–40 who had experienced a traumatic event in the last 2 years, completed 14-days of sleep diaries on which they also reported nightly presence of nightmares. Participants underwent an ambulatory polysomnography (PSG) acclimation night followed by a second PSG night from which sleep physiological measures were computed. PTSD severity was measured using the PTSD Checklist for DSM-5 (PCL-5). Dependent variables were total PCL-5 score as well as its hyperarousal symptom subscore. Predictors included REM latency, percent, density, segment length and an index of parasympathetic tone (root mean square of the successive differences in the R-R interval or RMSSD). Hierarchical regression models were conducted to analyze the association of REM features with PCL-5 total and hyperarousal subscales. Using hierarchical regression, REM-sleep RMSSD accounted for a significant proportion of the variation in outcome variables, even when

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Author Contribution

CD, KIO, JRSN and EFP authored the manuscript, CD conducted statistical analysis, TM, JS, and SPO consulted on data analyses, JPD adapted analysis software, KG scored sleep records, NBL interviewed participants, EFP supervised the manuscript.

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accounting for other REM-sleep features. The present findings support hypothesized relationships between PTSD symptomatology and REM-sleep physiology and, specifically, that lowered parasympathetic tone in REM may be an important associate of the hyperarousal symptom cluster in PTSD.

Keywords

REM sleep; Heart Rate Variability; Posttraumatic Stress Disorder; Parasympathetic Nervous System; Hyperarousal

Introduction

Approximately half of all people in the United States report having experienced at least one traumatic event, with 8% of women and 3.4% of men developing posttraumatic stress disorder (PTSD) at some point in their lives (Lehavot, Katon, Chen, Fortney, & Simpson, 2018). Individuals diagnosed with PTSD often experience symptoms of hyperarousal, sleep disruptions, and abnormalities in autonomic nervous system (ANS) function (Bertram et al., 2014). Studies have shown that, on average, patients with PTSD demonstrate increased heart rate (HR) both during wakefulness (Bertram et al., 2014) and at rest (Bertram et al., 2014); (Kobayashi, Lavela, Bell, & Mellman, 2016).

Observing these ANS differences, several PTSD studies have evaluated variability of the R-R interval in the electrocardiogram (ECG), i.e., heart rate variability (HRV), during wakefulness and sleep. Relative changes in the sympathetic (SNS) and parasympathetic (PNS) branches of the ANS have been demonstrated across these two behavioral states (Chouchou & Desseilles, 2014). Studies have focused on the strength of PNS outflow as indexed by the root mean square of the successive differences in the R-R interval (RMSSD) or by the absolute power of the high frequency band (0.15–0.40 Hz) of variations in the R-R interval (HF-power). These two, highly correlated variables reflect modulation of heart rate by the vagus nerve, the chief pathway of PNS outflow to the periphery from brainstem nuclei (Laborde, Mosley, & Thayer, 2017; Shaffer, McCraty, & Zerr, 2014). Two recent meta-analyses evaluating HRV found that individuals with PTSD showed decreased daytime RMSSD and HF-power compared to healthy controls (Ge, Yuan, Li, & Zhang, 2020; Schneider & Schwerdtfeger, 2020). Park and colleagues (Park et al., 2017) reported that individuals with PTSD from combat-related trauma exhibited decreased waking RMSSD and HF-power compared to healthy controls, and that daytime HRV measures of PNS activity were negatively associated with hyperarousal symptoms. Further, a study by Mäder et al., (2021) conducted in a sample encompassing the participants the present study found that HR acceleration during a loud-tone task was associated with greater self-reported PTSD severity. In that sample, HR acceleration significantly differed between PTSD-diagnosed participants and Trauma-Exposed Controls (TEC), but HRV did not (Mäder et al., 2021). In contrast, D'Souza and colleagues (D'Souza et al., 2019) did not find an association between daytime HF-power and PTSD symptom severity. Recent studies investigating HRV report negative correlations between HF-power and PTSD severity (Rissling et al., 2016) during sleep, but not wakefulness. Ulmer and colleagues (Ulmer, Hall, Dennis, Beckham,

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& Germain, 2018) reported that trauma-exposed control subjects had greater HF-power, and therefore more parasympathetic activity, during non-rapid eye movement (NREM) sleep than veterans with PTSD. However, HF-power differences between groups were not observed in REM sleep.

In both healthy and clinical populations, REM sleep has been linked with emotion regulatory processes (Goldstein & Walker, 2014; Walker & van der Helm, 2009). In a meta-analysis, Kobayashi, Boarts, and Delahanty (2007), reported greater REM density (measured as number of rapid eye movements per unit time) in individuals with PTSD, while some studies also report reduced REM% in individuals with PTSD compared to resilient, trauma-exposed individuals (Zhang et al., 2019). A study evaluating current and chronic PTSD (Mellman, Kobayashi, Lavela, Wilson, & Hall Brown, 2014) found that REM sleep was reduced and more fragmented (shorter segments of REM sleep) in PTSD patients following recent trauma exposure (see also (Mellman, Pigeon, Nowell, & Nolan, 2007). Mellman and colleagues further showed that subjective insomnia, nightmare severity, REM fragmentation and autonomic changes in REM during the early aftermath of traumatic injury predicted later development of post-traumatic symptoms (Mellman, Knorr, Pigeon, Leiter, & Akay, 2004; Mellman et al., 2007). Together these findings indicate that abnormalities in sleep, and in particular REM sleep, are key for understanding the mechanisms of PTSD (for further review, see (Germain, 2013).

Whereas previous studies have explored interrelationships of PTSD symptoms with waking and sleep HRV, the current study will explore whether specific features of REM are associated with PTSD hyperarousal (Cluster E) symptoms of PTSD. This study adopts the Research Domain Criteria (RDoC) approach (Insel et al., 2010), examining dimensionally how REM features vary across individuals exhibiting a range of symptom severity from highly abnormal to minimally present. We hypothesize that parasympathetic activity during REM, as measured by HRV RMSSD, will account for a significant proportion of the variance in daytime PTSD hyperarousal symptoms, after accounting for various measures of REM sleep integrity (i.e., REM segment length, REM%, REM density and REM latency).

Methods

Participants

122 participants were recruited from the greater Boston area using online and posted advertisements as part of a larger project. Out of the starting sample, 32 subjects were excluded from the final analyses due missing REM characteristic data. Missing values resulted from PSG equipment failure, corrupted EEG data files, or files that contained too much noise to be scorable, resulting in a sample of 90 participants. Participants were 70% female and ranged in age from 18 to 40 years old (mean = 23.56; SD = 4.53). Further demographic information and subsample characteristics can be found in Table 1. Demographic information and characteristics for the full sample of this study can be found in Denis et al., (2021). All participants reported having experienced at least one DSM-5, Criterion-A traumatic event in the last 2 years, but not in the last month. Potential participants underwent a preliminary telephone screening, and those who qualified underwent full psychiatric and sleep-disorders interviews as well as a urine toxicology

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screen. An experienced psychodiagnostician assessed current and lifetime psychiatric disorders using the Structured Clinical Interview for DSM-IV for Non-patients (SCID 1/NP (First, 2007)) as well as PTSD symptoms using the Clinician-Administered PTSD Scale for DSM-5 (CAPS-5 (Weathers, 2013)) and the PTSD Checklist for DSM-5 (PCL-5 (Blevins, Weathers, Davis, Witte, & Domino, 2015)). PTSD diagnosis was categorically established by the diagnostic clinician in accordance with the clinical interview and assessment battery. In analyses of the hyperarousal (Cluster E) items of the PCL-5 (primary outcome), the single sleep item was excluded. The sleep-disorders interview was completed using the Pittsburg Structured Clinical Interview for Sleep Disorders (a widely used (Insana, Hall, Buysse, & Germain, 2013) but unpublished, in-house instrument). All participants provided written consent to participate in the study and were paid for their participation. All procedures were approved by the Partners Healthcare Institutional Review Board.

Participants were excluded if they met one or more of the following criteria: history of chronic childhood abuse or neglect; PTSD diagnosis preceding traumatic event indexed at study interview; history of a major neurological or medical illness; psychosis; autism spectrum or bipolar disorders; sleep disorder other than Insomnia Disorder; current or previous year substance abuse or dependence; positive urine toxicology screen for 11 substances of abuse; caffeine consumption exceeding 5 beverages/day; alcohol consumption exceeding 10 drinks/week; any Magnetic Resonance Imaging (MRI) contradictions. Participants who were on stable doses (minimum eight weeks) of antidepressants, were diagnosed with a mild Axis-I anxiety disorder, dysthymia or remitted Major Depressive Disorder were allowed in the study. Antidepressant use was categorized into a binary (yes/no) variable.

Procedure

All participants completed approximately 14-days of wrist actigraphy and sleep diaries using the Evening-Morning Sleep Questionnaire sleep diary (EMSQ) (Pace-Schott, Kaji, Stickgold, & Hobson, 1994) with an accompanying nightmare questionnaire. During this period, participants also underwent an acclimation/sleep-disorders screening night using ambulatory polysomnography (PSG). After this 2-week sleep-monitoring period, participants completed a second night of ambulatory PSG (“Baseline PSG”). Over the following 2 days, they then completed a 2-session fear conditioning and extinction protocol with 2 evening functional MRI (fMRI) sessions 24 hours apart with an intervening third (“memory-consolidation”) PSG night. For 12% of participants, the Baseline PSG night took place during the week following fMRI scans. Participants were asked to refrain from all alcohol and recreational substances while completing study protocol. They were also asked to abstain from caffeine and daytime naps on the day before their Baseline PSG as well as the subsequent two days during which they completed the fear conditioning and extinction protocol.

Sleep and Nightmare Diaries

In the evening prior to sleep, the EMSQ queried the prior day’s activities and the time at which each participant began to attempt sleep. The morning portion queried awakening time, subjective sleep onset latency (SOL), number and duration of nocturnal awakenings, and

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ratings of sleep quality. The nightmare questionnaire was completed in the morning and asked participants to recall if they had a dream the previous night and whether their dream was a nightmare, a bad dream, or another type of dream. Nightmares were operationally defined as bad or unpleasant dreams that caused awakening and their rates of occurrence were calculated on a per-total-diary basis. Sleep diary entries were also used to compute subjective SOL, TST (time in bed [TIB] minus SOL and wake time after sleep onset [WASO]) and sleep efficiency (SE; TST as a proportion of TIB) for each night during the 2-week sleep-monitoring period.

Ambulatory Polysomnography

Using the Somte-PSG (Compumedics USA, Charlotte, NC), electrodes were attached in the laboratory using a standard montage that included 6 electroencephalograph (EEG) channels (F3, F4, C3, C4, O1, O2) with reference to contralateral mastoids (A1, A2), 2 electrooculograph (EOG) channels (right and left outer canthi), 2 submental electromyograph (EMG) channels, and 2 electrocardiogram (ECG) channels (right clavicle and left 5th intercostal space). Signals were recorded at 256 Hz, using high (0.16 Hz) and low (102 Hz) pass filters. Once instrumented, participants returned home to sleep. During the acclimation/screening (first PSG) night, additional channels for pulse-oximeter, respiration transducer belts, nasal cannula and tibialis PLM sensors were added to screen for obstructive sleep apnea (OSA) and periodic limb movement disorder (PLMD). No participant met criteria for clinically significant OSA or PLMD. The Baseline PSG night provided the data analyzed in the current study. All sleep records were scored by an experienced, RSPGT research polysomnographer using American Academy of Sleep Medicine criteria and Compumedics Profusion 4.0 software.

PSG Sleep Variables

Sleep stages (N1-N3, REM) were computed as percent of total sleep time. Because of the multiple associations of REM with PTSD risk, diagnosis, and symptomatology (Germain, 2013; Kobayashi et al., 2007; Kobayashi et al., 2016; Mellman et al., 2004; Mellman et al., 2007; Pace-Schott, Germain, & Milad, 2015; Ulmer et al., 2018; Zhang et al., 2019) the following variables characterizing this sleep stage were specifically computed as predictors of PTSD symptoms:

REM percent (REM%) ––REM% was calculated as the total amount of minutes spent in REM sleep per the total amount of minutes asleep, multiplied by 100.

REM-onset latency (REM Latency) ––REM Latency was calculated as the number of minutes occurring after sleep onset before the first REM period of the night as computed from scored records.

REM Density ––REM density quantifies the number of rapid eye movements (REMs) per minute of REM sleep. REM density was calculated using an open source, validated automatic algorithm (Yetton et al., 2016). European Data Format (EDF) files containing EOG data were extracted from scored PSG files. Likewise, hypnograms of scored epochs were extracted containing locations and durations of each REM period within a PSG record.

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The algorithm outputs REM density for each REM segment analyzed. Weighted averages using the duration of each REM period were calculated using the period-by-period results from the automatic detector and reported as REM density.

REM segment length (REM segment) ––Extracted hypnograms also provided the durations of all REM periods and these were averaged for each subject. Following (Mellman et al., 2007) the mean segment length (REM segment) was used as an index of REM fragmentation, with a shorter mean REM segment representing greater REM fragmentation.

REM and SWS parasympathetic indices (RMSSD, HF-Power) ––Separate ECG segments of at least 5 continuous minutes or more were extracted, as EDF files, during REM and SWS segments from the scored PSG records and analyzed using Kubios HRV premium software (Kubios Oy, Kuopio, Finland). Within Kubios, these segments were cleaned to remove any artifacts or ectopic beats such as premature ventricular contractions. Segments with too many artifacts or noise were removed from the analysis. This study explored two HRV measures of parasympathetic activity: RMSSD and HF-power. RMSSD is a time-domain measure that evaluates the beat-to-beat variance of heart rate (Laborde et al., 2017; Shaffer & Ginsberg, 2017) and constituted our primary REM HRV predictor of PTSD symptomatology (see below). HF-power is correlated with respiratory influences on HRV, which is modulated by parasympathetic activity of the autonomic nervous system and is therefore a reflection of vagal tone (Laborde et al., 2017; Shaffer & Ginsberg, 2017). Greater HF-power or RMSSD both indicate greater parasympathetic activity, and are highly intercorrelated (r_s= .980, $p < .0001$ in the current sample). Once RMSSD and HF-power measures from each segment of REM were obtained, weighted averages were calculated using the duration of their respective segments in each sleep stage to calculate REM RMSSD and REM HF-power values for each subject. HF-power scores were transformed by the decadic logarithm to normalize their distribution (logHF-power).

Outcome and predictor variables

The primary outcome variable was the hyperarousal (Cluster E) PTSD symptom subscore on the PCL-5 (PCL-5 hyperarousal) excluding item 20– “Trouble falling or staying asleep”– (PCL-5 hyperarousal-NS). Three secondary outcomes were the Cluster E for the CAPS-5 Hyperarousal scale (including the sleep item) as well as overall PTSD severity as measured by the total PCL-5 and CAPS-5 scores excluding the hyperarousal subscales. Predictor variables and rationale for their use follow: Diminished objective and subjective sleep quality have been extensively reported as characteristic of PTSD (Germain, 2013; Kobayashi et al., 2007; Kobayashi, Huntley, Lavela, & Mellman, 2012; Zhang et al., 2019). Among PSG measures, we selected those REM parameters most often associated with PTSD symptoms. These included REM density (Kobayashi et al., 2007), and REM percent (Zhang et al., 2019). In addition, we included average REM segment length, which when shortened soon after trauma, has been shown to predict later PTSD symptoms (Mellman, Bustamante, Fins, Pigeon, & Nolan, 2002; Mellman et al., 2007) REM latency, although not consistently abnormal in PTSD, is believed to reflect REM pressure in depression (Kobayashi et al., 2007; Palagini, Baglioni, Ciapparelli, Gemignani, & Riemann, 2013; Zhang et al., 2019). HRV measures during REM were included because autonomic abnormalities during sleep

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have been demonstrated in PTSD (Mellman et al., 2004; Ulmer et al., 2018). RMSSD was used as the HRV index in the subsequent analyses. RMSSD was preferred over HF-power as some studies have demonstrated HF-power may be affected by breathing patterns especially at sampling rates below 500 Hz (Shaffer & Ginsberg, 2017).

Statistical Analysis

To examine the contribution of REM sleep physiology to daytime PTSD symptomatology, we conducted four hierarchical regression models using the above predictor variables. Model I predicted PCL-5 hyperarousal-NS score (primary outcome). Model II predicted total PCL-5 score excluding hyperarousal items (PCL-5 Total – NH). Two additional models predicting hyperarousal and PTSD severity indexed by the CAPS-5 questionnaire can be found in the supplement (Supplementary Tables 1-2). Each model consisted of three steps of variable entry: Step 1) entered demographic variables (age, gender, and months since trauma) as well as a categorical variable indicating antidepressant use (antidepressant status, yes/no). Step 2) entered REM sleep characteristics associated with PTSD symptom severity (REM%, REM density, REM segments, and REM latency). Finally, Step 3) entered REM HRV (REM RMSSD) into the model. With these analyses, our aim is to explore whether REM RMSSD accounts for a significantly greater proportion of the variance in the outcomes than the variance accounted for in the previous steps. This is reflected in the change of the R^2 value (ΔR^2). In hierarchical regression, it is possible to have an overall model that is statistically significant at all steps, but in which the change in R^2 is not significant. Therefore, we will only interpret the model at step 3. A second set of hierarchical regression models exploring the contribution of Slow Wave Sleep (SWS; N3) RMSSD to hyperarousal and PTSD severity were also conducted and can be found in the supplementary section (Supplementary Tables 3-6).

A visual inspection of residual plots suggested that the assumption of a linear relationship between the predictors and outcome variables was sufficiently met. Thus, we proceeded with the hierarchical regression models considering the large sample size. All variables met acceptable ranges of skewness and kurtosis. Variance Inflation Factors (VIFs) ranged between (1.014 and 1.189), indicating that the assumption of absence of multicollinearity was sufficiently met. Little's MCAR test indicated missing data were missing-completely-at-random ($\chi^2 = 286.84$, $p = 0.054$). Missing data were handled using pairwise deletion. All analyses were two-tailed and evaluated at an $\alpha = 0.05$ confidence level. Outliers were manually excluded using a cut-off of ±3 standard deviations. Since a Shapiro-Wilks test indicated non-normality of some REM-sleep variables, bivariate correlations between predictor and outcome variables were conducted using non-parametric Spearman regression. Outcome variables sufficiently met assumptions of normality. All statistical tests were conducted using SPSS (version 28.0.0).

Results

Demographic information and trauma-related characteristics of the sample can be found in Table 1. Overall sleep characteristics in the total sample and broken down by PTSD diagnosis status, viz. PTSD and Trauma Exposed Controls (TEC), are provided in Table 2.

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An independent samples t-test indicated significant gender differences for REM RMSSD (t(88) = 2.809, p = .006, Cohen's d = 0.65), with males demonstrating greater REM RMSSD (M = 81.82, SD = 38.98) than females (M = 59.73, SD = 31.97).

Spearman's rho correlations, shown in Table 3, indicate a positive correlation between PCL-5 Hyperarousal-NS and nightmare rates ($r_s(88) = 0.24$, p = .027). CAPS-5 total–NH scores were also positively correlated with nightmare rates ($r_s(88) = 0.29$, p = 0.002). There was a negative correlation between PCL-5 Hyperarousal-NS and REM RMSSD ($r_s(88) = -0.213$, p = .044). There were no significant correlations between PCL-5 Total - NH, or CAPS-5 Hyperarousal NS scores with other REM-sleep characteristics. We nonetheless included these variables in the hierarchical regression models based upon previous research suggesting their associations with PTSD symptomatology.

Model I. Hierarchical regression of REM sleep characteristics predicting PCL-5 Hyperarousal scores excluding sleep item.

Hierarchical regression model summaries for PCL-5 Hyperarousal-NS scores can be found in Table 4. Step 1 of the hierarchical regression analysis entered demographic variables and antidepressant status into the model. This step did not account for a significant proportion of the variance in PCL-5 hyperarousal-NS (Adj. R^2 = .032; $F_{(4, 83)} = 1.718$, p = .154). Step 2 entered the block of REM sleep features (REM%, REM latency, REM density, and REM segment) into the model. The model at step 2 contained demographic variables, antidepressant status, and all four REM sleep feature variables. This step did not account for a significant increase in the proportion of the variance in PCL-5 hyperarousal-NS accounted for by the model at step 1 (ΔR^2 = .062, p = .238). Step 3 entered REM RMSSD into the model. The model at step 3 included demographic variables, antidepressant status, all four REM sleep feature variables, and REM RMSSD. This step accounted for a significant increase in the proportion of the variance in PCL-5 hyperarousal-NS accounted for by the model at step 2 (ΔR^2 = .057, p = .022). These results suggest that REM RMSSD accounts for a significant proportion of the variation in hyperarousal, above and beyond that accounted for by REM sleep features, demographic variables, and antidepressant status. A summary of this analysis can be found in Table 4.

The final model at step 3, containing all nine predictor variables, explained a significant proportion of the variance in PCL-5 Hyperarousal-NS scores (Adj. R^2 = .102, $F_{(9, 78)}$ = 2.096, p = .040). Results indicate that for every one standard deviation decrease in REM RMSSD ms^2, which indexes parasympathetic nervous system activity, there is a corresponding 0.265 standard deviation increase in wake hyperarousal symptoms reported ($t_{(78)} = -2.344$, p = .022, CI = [−0.052, −0.004]; Cohen's $f^2 = 0.07$) when all other variables are held constant, although the effect is small. REM density was also a trending predictor of PCL-5 Hyperarousal-NS scores in the final model, such that for every one standard deviation decrease in REM density there is a 0.248 standard deviation increase in hyperarousal symptoms ($t_{(87)} = -2.166$, p = .033, CI = [−0.042, −0.017]), when all other variables are held constant.

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Model II. Hierarchical regression of REM sleep characteristics predicting PCL-5 Total - NH scores.

Hierarchical regression and ANOVA model summaries predicting PCL-5 Total – NH can be found in Table 5. As in Model I, hierarchical regression at step 1 entered demographic variables and antidepressant status into the model. This step did not account for a significant proportion of the variance in PCL-5 scores (Adj. R^2 = .015; $F_{(4, 83)}$ = 1.341, p = .262). Step 2 entered the block of REM sleep features into the model (REM%, REM density, REM segment, and REM latency). This step did not account for a significant increase in the proportion of variance in PCL-5 Total- NH scores compared to step 1 (ΔR^2 = .030, p = .635). In step 3, REM RMSSD was added to the model. This step also did not account for a significant proportion of the variance in PCL-5 Total- NH scores compared to step 2 (ΔR^2 = .014, p = .270). The final model containing all nine predictor variables did not explain a significant proportion of the total variance in PCL-5 Total-NH scores (Adj. R^2 = .001; $F_{(9, 78)}$ = 1.010, p = .439).

Results for the hierarchical regression models exploring the contribution of REM RMSSD to CAPS-5 hyperarousal and total CAPS-5 score (excluding hyperarousal) can be found in the supplement (Supplementary Tables 1-2). While the final model explained significant proportion of the variance in both CAPS-5 hyperarousal and CAPS-5 total scores, REM RMSSD did not contribute to a significant proportion of the variance in either model when all other REM characteristics were accounted for. The model exploring the contribution of SWS RMSSD to PCL-5 hyperarousal scores, show that SWS RMSSD does not account for a significant increase in the proportion of the variance accounted for by the model beyond that accounted for by demographic variables, antidepressant status, and SWS percentage (ΔR^2 < .001, p = .931). The final model containing all eight predictors did not explain a significant proportion of the variance in PCL-5 hyperarousal scores (Adj. R^2 = .006; $F_{(6, 78)}$ = 1.081, p = .381). The same was true for the model where SWS predicted PCL-5 total score (ΔR^2 < .001, p = .841; Adj. R^2 = .004, $F_{(6, 78)}$ = 1.052, p = .399). Results for SWS hierarchical regression models can be found in the supplement (Supplementary Tables 3-6).

Discussion

We investigated the relationship of PTSD hyperarousal symptoms with REM-sleep variables in a large sample of individuals exposed to psychological trauma within the past 2 years. Approximately half of these individuals were diagnosed with PTSD. Adopting the RDoC approach (Insel et al., 2010), hierarchical regression models indicated that a REM-sleep parasympathetic index, RMSSD, was a significant negative associate of the waking hyperarousal (Criterion E) symptoms of PTSD (PCL-5 Hyperarousal-NS) across the spectrum of trauma sequela in a sample that included PTSD-diagnosed, sub-clinical, and resilient individuals. Importantly, this association was seen while controlling for other REM sleep features reported to be associated with PTSD risk including reduced average REM period length (REM segment) (Mellman et al., 2002; Mellman et al., 2007) increased REM density (Kobayashi et al., 2007), and decreased REM percent of total sleep time (Zhang et al., 2019) (see Table 4). Among these same REM-sleep features, HRV indices alone significantly differentiated individuals diagnosed with PTSD and trauma-exposed

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individuals without PTSD (see Table 2). In addition, REM RMSSD correlated negatively with REM density, a variable believed to reflect forebrain arousal in REM sleep (Paul, Alpers, Reinhard, & Schredl, 2019) that has also been shown to be elevated in a meta-analysis of sleep disturbances in PTSD (Kobayashi et al., 2007).

However, results indicate that REM RMSSD did not account for an incremental proportion of the variance in overall PTSD severity, demonstrated by PCL-5 and CAPS-5 total scores not including hyperarousal, once other REM features were entered into the model (see Table 5 and Supplementary Table 2, respectively). REM RMSSD also did not account for a significant incremental proportion of the variance in the cluster E symptoms captured by the CAPS-5 Hyperarousal subscale (Supplementary Table 1). The four SWS models showed that SWS RMSSD did not account for a significant increase in the proportion of the variance in PTSD hyperarousal or severity when demographic variables, antidepressant status, and SWS percentage were included in the model (Supplementary Tables 3-6). While these supplemental models shed some light on the relationship of SWS HRV on self-reported hyperarousal symptoms, the lack of equivalent SWS characteristics make them unsuitable for full comparison against the REM models. However, Figure 1 suggests that there may be a general effect of HRV in both REM and SWS stages in self-reported hyperarousal.

Many studies on waking HRV show reduced PNS activity in PTSD compared to controls. Indeed, reduced PNS activity has been suggested to be an endophenotype of PTSD (Ge et al., 2020). In the current sample, however, HRV measures in wakefulness did not differentiate between those diagnosed with PTSD and trauma-exposed controls (Mäder et al., 2021). The current study, however, shows lower PNS activity in trauma-exposed individuals with versus without PTSD during REM sleep (Table 2). REM sleep, like waking, is a brain-activated state in which emotional processing may be occurring (Goldstein & Walker, 2014). Our results further suggest that lower levels of PNS activity during REM may be an important associate of perceived PTSD hyperarousal symptoms across the spectrum of such symptoms in a trauma-exposed population. Thus reduced PNS activity during REM may constitute a specific biomarker of the hyperarousal cluster of PTSD symptoms.

Comprehensive views of PNS function, including Neurovisceral Integration and Polyvagal theory, view PNS activity not only as a means to regulate cardiovascular activity but also as an index of the ability to regulate emotional arousal in the face of stress (Smith, Deits-Lebehn, Williams, Baucom, & Uchino, 2020). Thus, lowered PNS activity during REM, a sleep stage widely hypothesized to be of importance to offline emotional processing (Goldstein & Walker, 2014; Pace-Schott et al., 2015), may be associated with the abnormal emotion regulation and reduced psychological resilience that increases risk for PTSD following a traumatic event (Kobayashi, Lavela, & Mellman, 2014). The gender differences in REM RMSSD observed in this sample, where males demonstrate greater REM RMSSD than females, likely reflect this generally greater resilience to stress in individuals with greater RMSSD (Smith et al., 2020). This is evidenced by the overwhelmingly greater number of males in the TEC group compared to the PTSD group (see Table 1).

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In addition to hyperarousal symptoms, a possible indirect associate of the re-experiencing symptom cluster of PTSD was suggested by a negative correlation between RMSSD and REM density. REM density, in turn, trended towards a significant correlation with the rate of nightmare occurrence (Spearman's Rho = 0.188, $p = 0.080$). Associations between REM density and nightmare occurrence was also found by Paul et al., (2019). Notably, psychophysiological investigations suggest that rapid withdrawal of PNS braking of heart rate in response to a startling stimulus during wakefulness may be associated with a vulnerability to nightmares in trauma-exposed individuals (Mäder et al., 2021; Tanev et al., 2017). Both a decline in PNS activity and an increased vulnerability to nightmares may result from elevations in noradrenergic activity that has been suggested to occur during sleep in PTSD (reviewed in (Pace-Schott et al., 2015; Richards, Kanady, & Neylan, 2020).

There are several limitations to the current study. Although a prior study found reduced PNS activity in PTSD versus TEC in NREM but not REM sleep, their findings were in a differing population (veterans) (Ulmer et al., 2018). The current civilian sample of otherwise healthy young adults show relatively low maximum and mean severity of PTSD symptoms whether or not an individual met CAPS-5 criteria for PTSD diagnosis (Table 1). Additionally, the current sample is heavily biased toward younger females. Thus, these findings may not generalize well to more severe PTSD presentations such as seen in some combat-exposed veterans. This imbalance within the sample also limits our ability to interpret the contribution of gender to PNS activation during REM. Further, whereas RMSSD has been demonstrated to be a reasonable indicator of PNS activation (Ulmer et al., 2018), there may also be sympathetic activation in PTSD that was not measured in the current study. Additionally, a power analysis (conducted in G*Power 3.1.9; Faul et al., 2007) indicated the present study had 84% power to detect a small effect size (Cohen's $f^2 = 0.1$). However, the effect size observed for the incremental increase in R^2 for REM RMSSD was much smaller (Cohen's $f^2 = 0.07$). An analysis of the sample's power to detect such a small change in R^2 was 69%, making it plausible that we lacked the power to detect the incremental increase in the proportion of variance in CAPS-5 hyperarousal and possibly overall PTSD severity measures by REM RMSSD. Therefore, while we can suggest that REM parasympathetic activity is reduced in individuals with greater hyperarousal symptoms, we cannot state that sympathetic activation doesn't also increase in more symptomatic participants during REM sleep. Furthermore, REM parameters, including RMSSD, were averaged across an entire night of sleep. Although sleep stage is believed to predominate over circadian influence in determining HRV (Viola et al., 2002), HRV parameters within a sleep stage are known to change across a night of sleep in PTSD (Kobayashi et al., 2016; Ulmer et al., 2018). Thus, the influence of time of night and sleep cycle on HRV could be investigated in future analyses. Lastly, unlike Ulmer et al. (2018), slow wave sleep (SWS) rather than NREM sleep as a whole was investigated in the current study.

Conclusions

Results support an often-hypothesized relationship of PTSD symptoms with abnormal REM sleep physiology that has been described in EOG and EEG biosignals as well as in HRV measures. Specifically, parasympathetic activity in REM decreased with increasing PTSD

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hyperarousal symptoms in a large sample of trauma-exposed individuals spanning the normal to pathological spectrum of posttraumatic symptomatology.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability Statement

The data underlying this article will become available in the future in the NIMH Data Archive (NDA) at https://nda.nih.gov, and can be accessed following instructions at https://nda.nih.gov/get/access-data.html.

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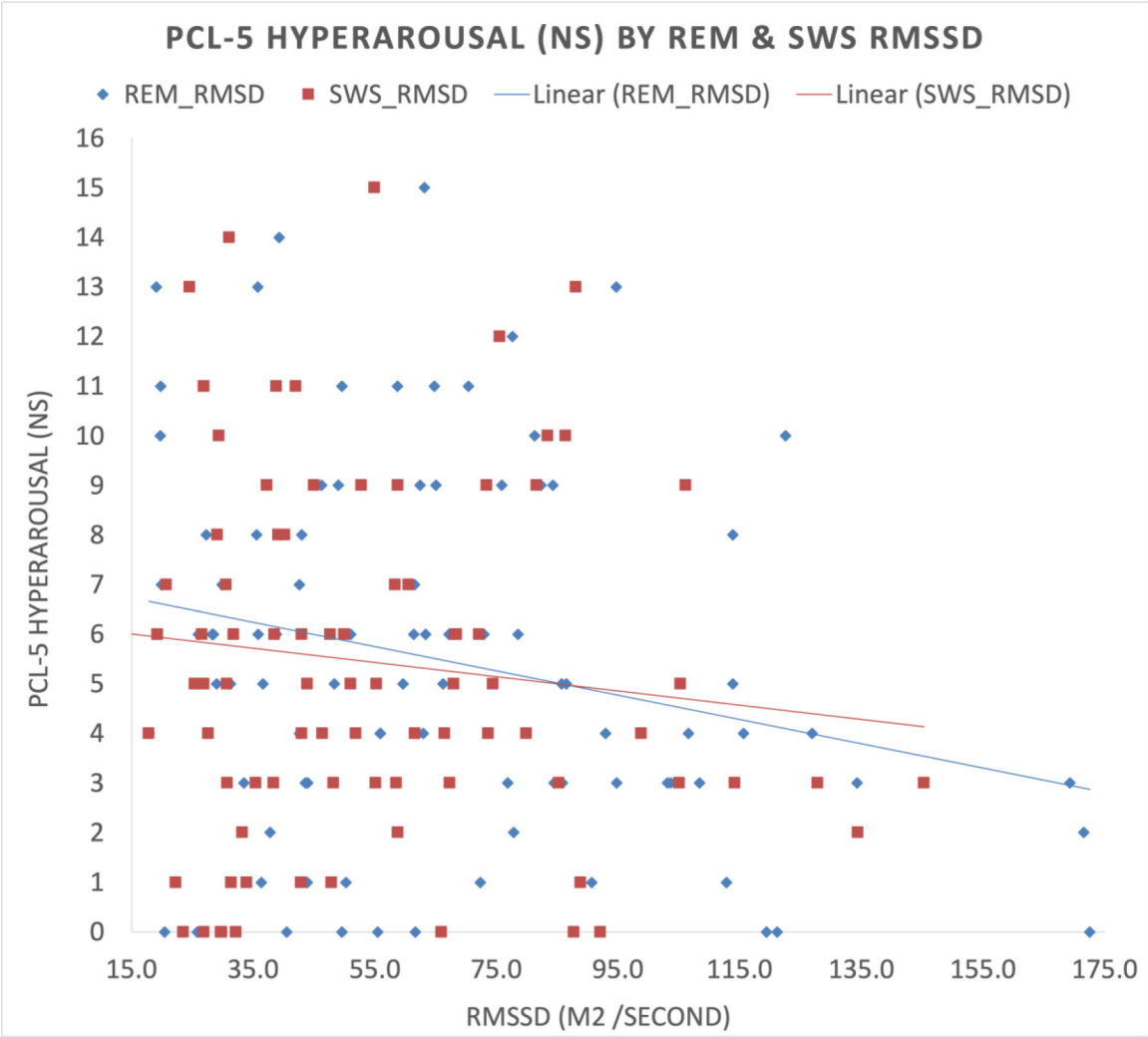

Figure 1.
Plot of PCL-5 Hyperarousal (excluding the sleep item) against REM and SWS RMSSD.

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Table 1.

Demographic and Trauma-related Sample Characteristics

	PTSD Mean ± SD	TEC Mean ± SD	Sample Mean ± SD
Sample Size	N = 42	N = 48	N = 90
Age (years)	24.26 ± 5.10	22.94 ± 3.92	23.56 ± 4.53
Months Since Trauma	13.83 ± 5.70	12.23 ± 7.24	12.98 ± 6.58
Antidepressant Status	Yes: 11 No: 30 Unknown: 1	Yes: 6 No: 42 Unknown: 0	Yes: 20 No: 78 Unknown: 1
PTSD Severity			
CAPS-5 Total- NH	22.10 ± 5.62	7.79 ± 4.80	14.47 ± 8.84
CAPS-5 Hyperarousal	8.88 ± 2.84	2.73 ± 2.47	5.60 ± 4.06
PCL-5 Total - NH	29.21 ± 9.63	14.04 ± 8.47	21.12± 11.77
PCL-5 Hyperarousal-NS	7.69 ± 3.22	3.54± 3.02	5.48 ± 3.73
Gender			
	Female: 36 Male: 6	Female: 27 Male: 21	Female: 63 Male: 27
Race			
American Indian or Alaskan Native: 2.2%		Asian: 10%	
Black or African American: 16.6%		White: 60%	
More than one race: 10%		Unknown or Not Reported: 1.1%	
Ethnicity			
Hispanic or Latino: 13.3%			
Type of Trauma			
Violent Assault: 17.8%		Rape or Sexual Assault: 18.8%	
Mass Shooting: 4.4%		Severe Human Suffering: 12.1%	
Sudden loss of family or friend: 5.5%		Transportation Accident: 33.3%	
Other: 8.8%			

Sample Characteristics. Mean and standard deviation for the Clinician Administered PTSD Scale for DSM-5 Total excluding hyperarousal subscale (CAPS-5 Total - NH), Clinician Administered PTSD Scale for DSM-5 hyperarousal subscale (Cluster E; CAPS-5 Hyperarousal), PTSD Checklist for DSM-5 Total excluding hyperarousal subscale and sleep item (PCL-5 Total - NH), PTSD Checklist for DSM-5 hyperarousal subscale excluding the sleep item (PCL-5 Hyperarousal - NS). Percentages across participants' sex, race and type of trauma are provided.

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Table 2.

Descriptive Sleep Characteristics

	Sample Size	Full Sample Mean ± SD	TEC Mean ± SD N = 48	PTSD Mean ± SD N = 42	*t*-stat (df)	*p*-value
REM Latency	89	98.27 ± 47.46	104.25 ± 52.92	91.44 ± 39.82	*t*(88) = 1.282	0.203
REM Percentage	90	18.15 ± 6.12	18.08 ± 4.99	18.23 ± 7.27	*t*(88) = −0.111	0.912
REM Density	90	7.10 ± 4.31	6.88 ± 4.02	7.35 ± 4.65	*t*(88) = −0.519	0.605
REM Segment Length (min)	90	8.43 ± 3.95	8.19 ± 3.36	8.72 ± 4.58	*t*(87) = −0.617	0.539
REM RMSSD	90	66.35 ± 35.49	**73.98 ± 40.73**	**57.64 ± 26.19**	***t*(88) = 2.291**	**0.012** *
REM HF-power log	88	3.03 ± 0.45	3.08 ± 0.48	2.96 ± 0.41	*t*(86) = 1.296	0.099
N1 Percentage	89	5.96 ± 3.44	6.22 ± 3.57	5.66 ± 3.31	*t*(87) = 0.766	0.223
N2 Percentage	89	56.29 ± 9.38	56.77 ± 7.89	55.76 ± 10.64	*t*(87) = 0.505	0.268
N3 Percentage	90	19.67 ± 10.73	19.08 ± 9.97	20.34 ± 11.62	*t*(88) = −0.555	0.290
SWS RMSSD	**86**	**54.41 ± 28.78**	**60.69 ± 33.39**	**46.83 ± 21.78**	***t*(84) = 2.360**	**0.010** **
SWS HF-power log	**86**	**2.90 ± 0.44**	**2.99 ± 0.45**	**2.80 ± 0.41**	***t*(84) = 2.039**	**0.022** *

Means, standard deviations, and t-test mean comparisons between PTSD and trauma exposed controls (TEC) sleep variables; Polysomnography REM Latency, REM Density, REM Segment Length, REM Root Mean Squared of the Successive Differences (REM RMSSD), $\log_{10}$ of REM High-Frequency Power (REM HF-power log), Slow Wave Sleep Root Mean Squared of Successive Differences (SWS RMSSD), Slow Wave Sleep $\log_{10}$ High-frequency Power (SWS HF-power log).

* p -value > 0.05

** p -value > 0.01.

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Table 3.

Spearman Correlation Table

	MONTHS ST	CAPS-5 TOTAL NH	CAPS-5 HYPE	PCL-5 HYPE-NS	PCL-5 TOTAL-NH	NIGHTMARE RATE	REM LATENCY	REM%	REM DENSITY	REM SEGMENT	REM RMSSD	REM LOG HF-POWER	N1%	N2%	N3%	SWS RMSS	SWS LOG HF-POWER
MONTHS ST	--																
CAPS-5 TOTAL-NH	0.095	--															
CAPS-5 HYPE	0.125	0.785**	--														
PCL-5 HYPE-NS	0.034	0.658**	0.684**	--													
PCL-5 TOTAL-NH	0.045	0.798**	0.649**	0.690**	--												
NIGHTMARE RATE	0.043	**0.290****	0.221*	0.182	**0.236***	--											
REM LATENCY	−0.025	−0.04	−0.157	−0.12	−0.079	−0.04	--										
REM%	0.232*	0.076	0.027	0.049	0.04	0.332**	−0.304**	--									
REM DENSITY	0.038	0.047	0.027	−0.08	−0.01	0.188	−0.037	0.168	--								
REM SEGMENT	0.126	0	−0.179	−0.079	−0.067	−0.066	0.124	0.289**	0.138	--							
REM RMSSD	0.02	−0.13	−0.134	**−0.213***	−0.13	−0.059	**−0.210***	0.05	−0.358**	−0.015	--						
REM LOG HF-POWER	0.006	−0.105	−0.11	−0.162	−0.09	−0.04	−0.172	0.04	−0.351**	−0.055	0.980**	--					
N1%	**0.225***	−0.062	0.014	0.004	−0.06	0.024	−0.199	0.155	−0.052	−0.393**	−0.011	0.008	--				
N2%	0.105	−0.148	−0.013	−0.046	−0.098	−0.273*	0.230*	−0.306**	0.032	−0.184	−0.061	−0.074	0.128	--			
N3%	**−0.267***	0.047	−0.026	−0.042	0.017	0.023	0.057	−0.325**	−0.081	0.099	0.025	0.034	−0.543**	−0.658**	--		
SWS RMSSD	−0.136	−0.172	−0.151	−0.091	−0.05	−0.119	−0.219*	−0.119	−0.425**	0.032	0.830**	0.810**	−0.144	−0.095	0.166	--	
SWS LOG HF-POWER	−0.087	−0.154	−0.15	−0.068	−0.042	−0.089	−0.172	−0.112	−0.407**	0.025	0.795**	0.802**	−0.14	−0.081	0.148	0.973**	--

Spearman correlations (two-tailed) among Months Since Trauma (MONTHS ST), Clinician Administered PTSD Scale for DSM-5 (CAPS-5), PTSD Checklist for DSM-5 (PCL-5), PTSD Checklist for DSM-5 Hyperarousal subscale (PCL-5 HYPE), PTSD Checklist for DSM-5 Hyperarousal subscale without sleep items (PCL-5 HYPE-NS), Nightmare Rates (NIGHTMARE RATE), Percentage of Rapid Eye Movement (REM%), REM sleep density (REM DENSITY), REM Segment Length (REM SEGMENT), REM sleep Root Mean Squared of Successive Differences (REM RMSSD), Log of REM sleep

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High-Frequency Power (REM LOG HF-POWER), Percentage of N3 Sleep Stage (N3%), Slow Wave Sleep Root Mean Squared of Successive Differences (SWS RMSSD), Log of Slow Wave Sleep High Frequency Power (SWS LOG HF-POWER).

*p-value < 0.05

**p-value < 0.01.

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Table 4.

Regression Steps and ANOVA Summaries of Hierarchical Model Predicting PCL-5 Hyperarousal-NS Scores

Step and Predictor Variable	B	β	CI	Adj. R^2	Δ R^2
Step 1:					
Age	0.014	0.017	[−0.166, 0.195]	0.032	0.076
Gender	1.728	0.213	[−0.113, 3.569]		
Month ST	−0.041	−0.073	[−0.169, 0.086]		
Antidepressant Status	2.025	0.215	[−0.225, 4.276]		
Step 2:					
REM %	−0.001	−0.001	[−0.15, 0.149]	0.051	0.062
REM Latency	−0.015	−0.196	[−0.034, 0.003]		
REM Density	**−0.215**	**−0.248**	**[−0.412, −0.017]**		
REM Segment	−0.103	−0.11	[−0.314, 0.107]		
Step 3:					
REM RMSSD	**−0.028**	**−0.265**	**[−0.052, −0.004]**	**0.102** **	**0.057** **

ANOVA Model Summary - PCL-5 Hyperarousal-NS

Model	SS	Df	Mean Square	F	Sig
1 Regression	92.564	4	23.141	1.718	.154
Residual	1118.061	83	13.471		
Total	1210.625	87			
2 Regression	167.071	8	20.884	1.581	.144
Residual	1043.554	79	13.21		
Total	1210.625	87			
3 Regression	235.747	9	26.194	**2.096**	**.040**
Residual	974.878	78	12.498		
Total	1210.625	87			

Demonstrates the model summary for all three steps of the hierarchical regression model predicting PCL-5 hyperarousal-NS scores as well as the ANOVA model comparisons for each of the three steps of the hierarchical regression model. RMSSD = Root Mean Squared of Successive Differences; Months ST =. Months Since Trauma; Antidepressant Status is a binary variable (yes/no).

** p-value <0.05.

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Table 5.

Regression Steps and ANOVA Summaries of Hierarchical Model Predicting PCL-5 Total NH Scores

Step and Predictor Variable	B	β	CI	Adj. R^2	Δ R^2
Step 1:					
Age	0.301	0.116	−0.300, 0.901	0.015	0.061
Gender	3.677	0.144	−2.449, 9.803		
Months ST	−0.146	−0.082	−0.571, 0.278		
Antidepressant Status	5.758	0.193	−1.731, 13.248		
Step 2:					
REM %	−0.01	−0.005	−0.508, 0.488	0.002	0.030
REM Latency	−0.014	−0.056	−0.074, 0.046		
REM Density	−0.291	−0.107	−0.948, 0.366		
REM Segment	−0.448	−0.15	−1.147, 0.251		
Step 3:					
REM RMSSD	−0.044	−0.132	[−0.123, −0.035]	0.001	0.014

ANOVA Model Summary - PCL-5 Total - NH

Model	SS	Df	Mean Square	F	Sig
1 Regression	731.979	4	182.995	1.341	.262
Residual	11324.515	83	136.44		
Total	12056.495	87			
2 Regression	1088.001	8	136	0.980	.458
Residual	10968.494	79	138.842		
Total	12056.495	87			
3 Regression	1258.594	9	139.844	1.010	.439
Residual	10797.901	78	138.435		
Total	12056.495	87			

Demonstrates the model summary for all three steps of the hierarchical regression model predicting PCL-5 Total scores excluding the hyperarousal subscale (PCL-5 Total – NH) as well as the ANOVA model comparisons for each of the three steps of the hierarchical regression model. RMSSD = Root Mean Squared of Successive Differences; Months ST =. Months Since Trauma; Antidepressant Status is a binary variable (yes/no).

** p-value <0.05.